

Exploring Genes

Stryer ch 5

- Basic tools of gene exploration
- Recombinant DNA technology
- Manipulating genes in eukaryotes
- Engineering novel proteins by site-specific mutagenesis

Recombinant DNA technology has revolutionized all aspects of biology. It is based on nucleic acid enzymology

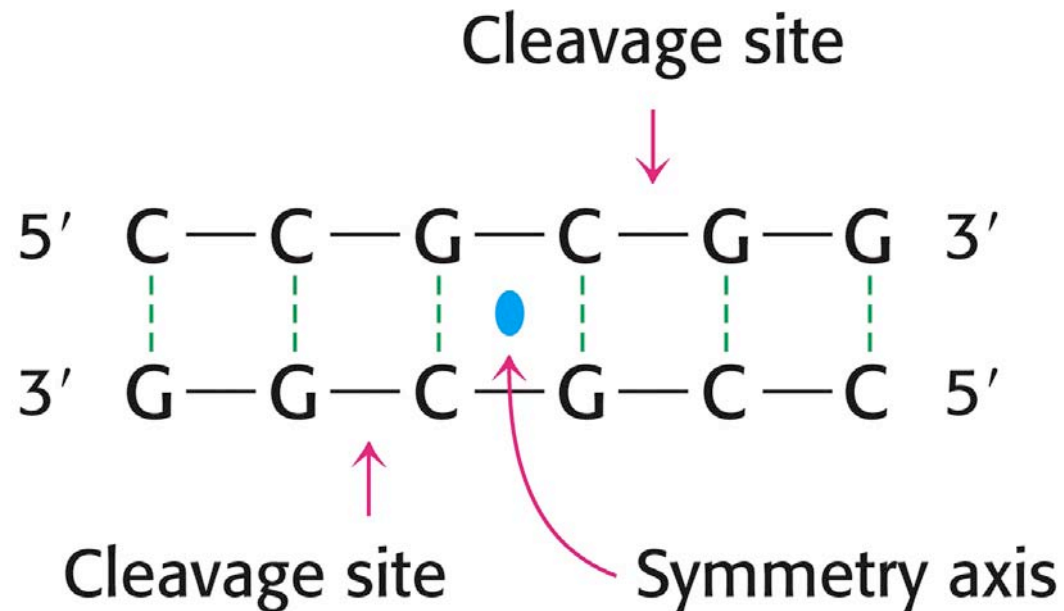
- Pioneered by Paul Berg, Henry Boyer, Stanley Cohen, Werner Arber around 1970
- Needs enzymes that cut, join, & replicate DNA and reverse transcribe RNA
- Restriction enzymes cut very long DNA molecules into specific, handable pieces
- DNA ligases join fragments together

Basic tools of gene exploration

- Restriction enzymes
- Blotting techniques
- DNA sequencing
- Solid phase synthesis of nucleic acids
- Polymerase chain reaction (PCR)

Restriction enzymes split DNA into specific fragments

Werner Arber, Hamilton Smith, Daniel Nathans



In each strand, the enzyme cleaves the C-G phosphodiester bond on the 3'side of the symmetry axis

Specificities of some restriction endonucleases

Recognize specific base sequences in double-helical DNA

Cleave both strands of duplex at specific places

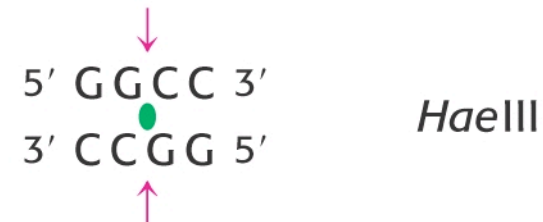
Application for:

Analyzing chromosome structure

Sequencing long DNA molecules

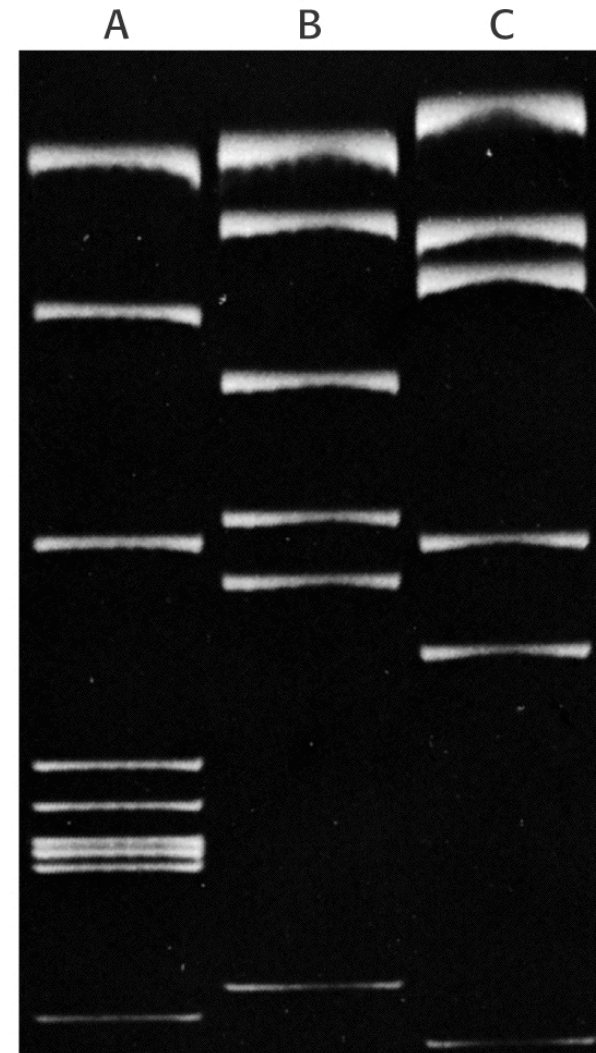
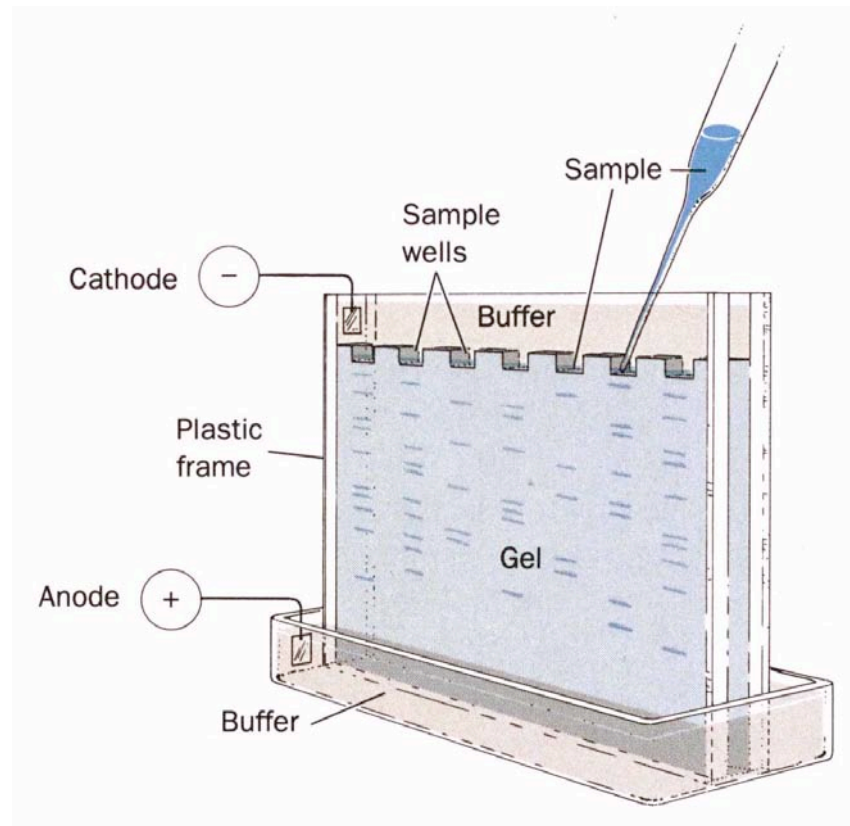
Isolating genes

Creating new DNA molecules



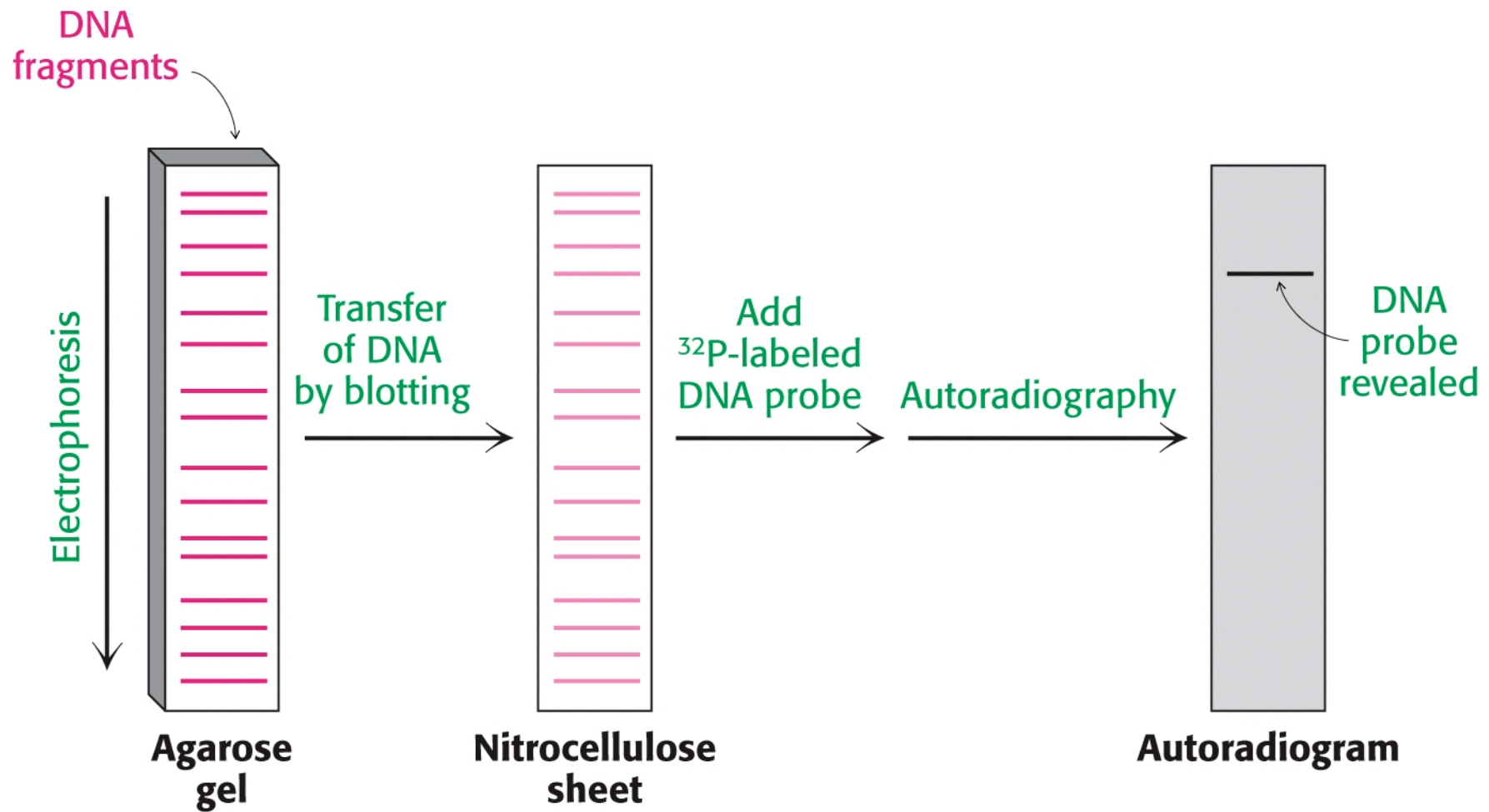
Gel electrophoresis pattern of a restriction enzyme digest

Small differences between DNA molecules can be distinguished (ca one nucleotide of hundreds)

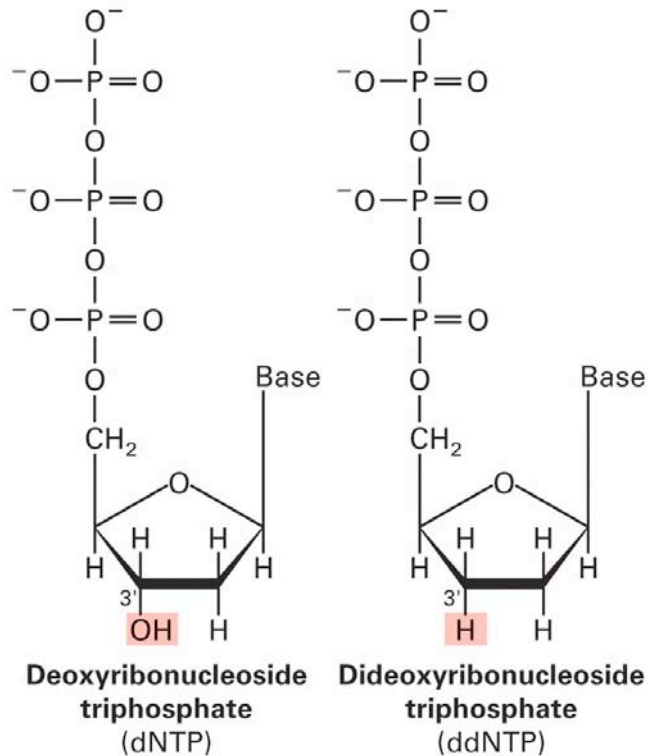


Fragments by 3 different enzymes

Southern blotting: A restriction fragment is identified by hybridization with a labeled complementary DNA strand

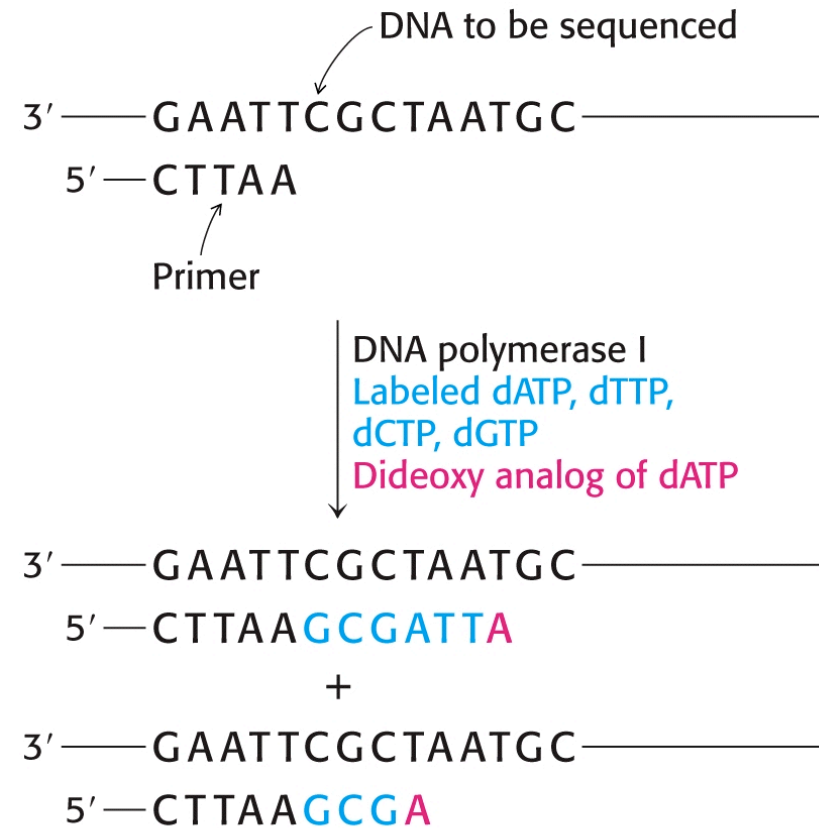


Sanger dideoxy method: DNA is sequenced by controlled termination of replication



dATP*, dTTP*
dCTP*, dGTP*

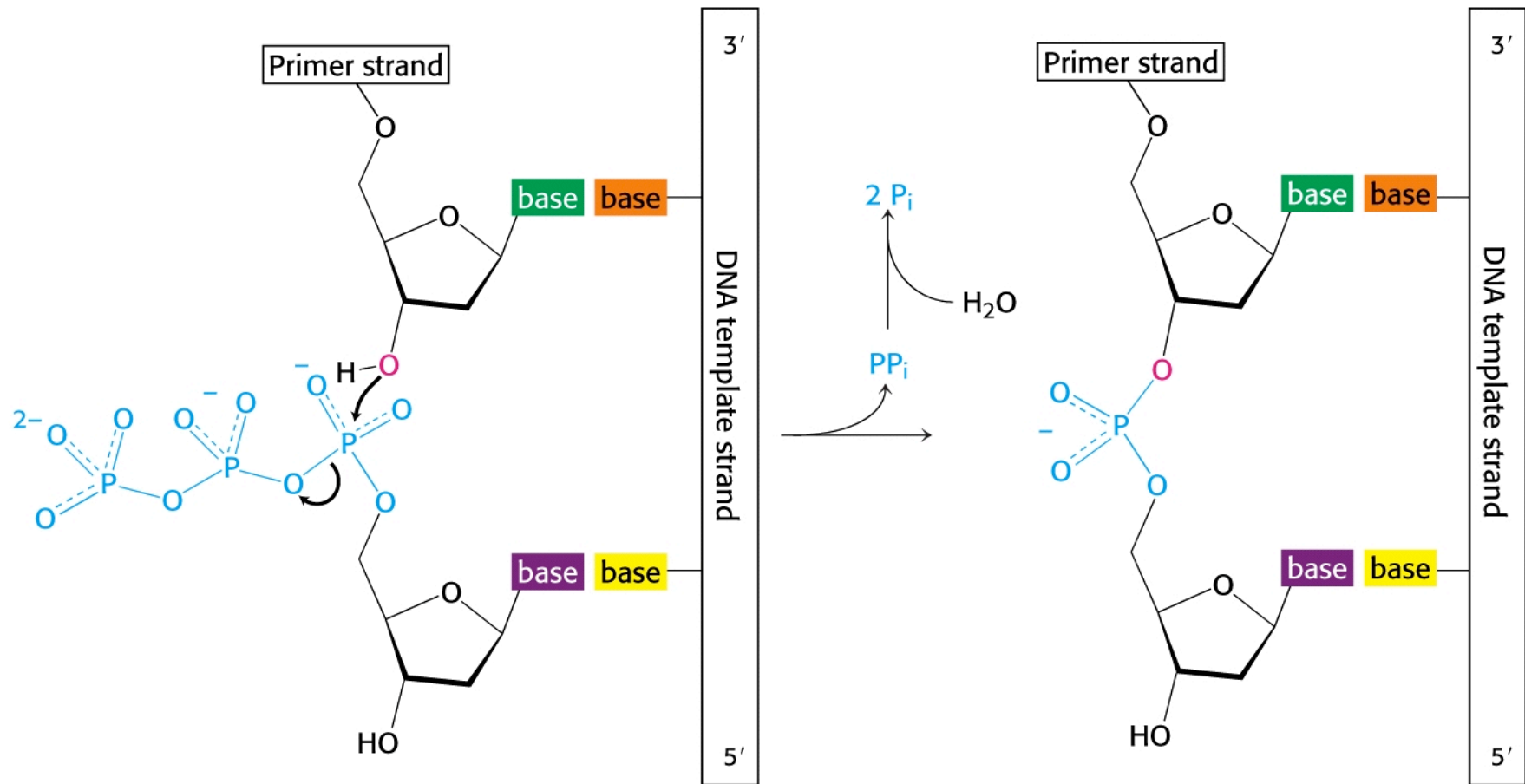
ddATP



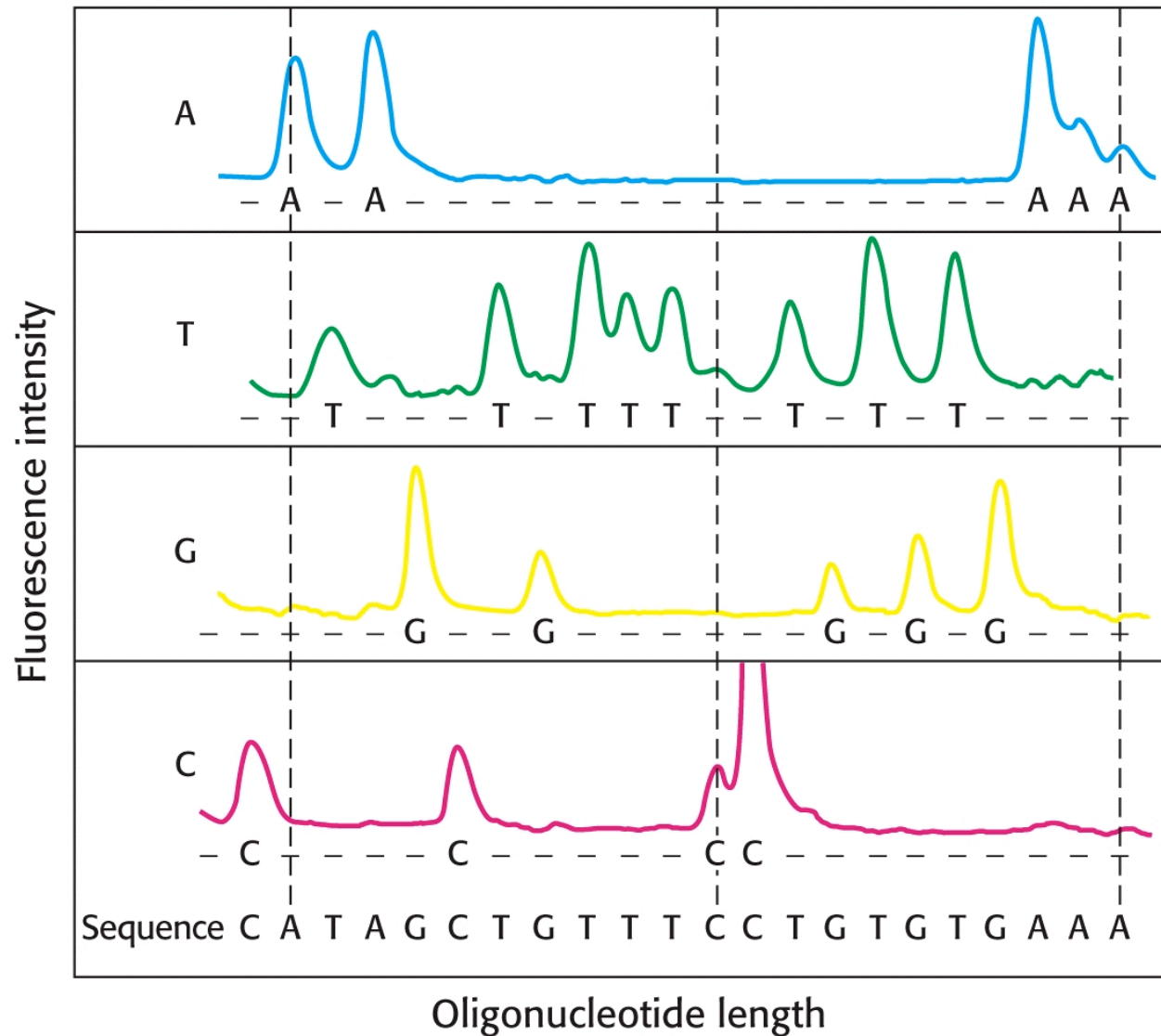
New DNA strands are separated
and electrophoresed

DNA replication by DNA polymerase

(a template directed enzyme)

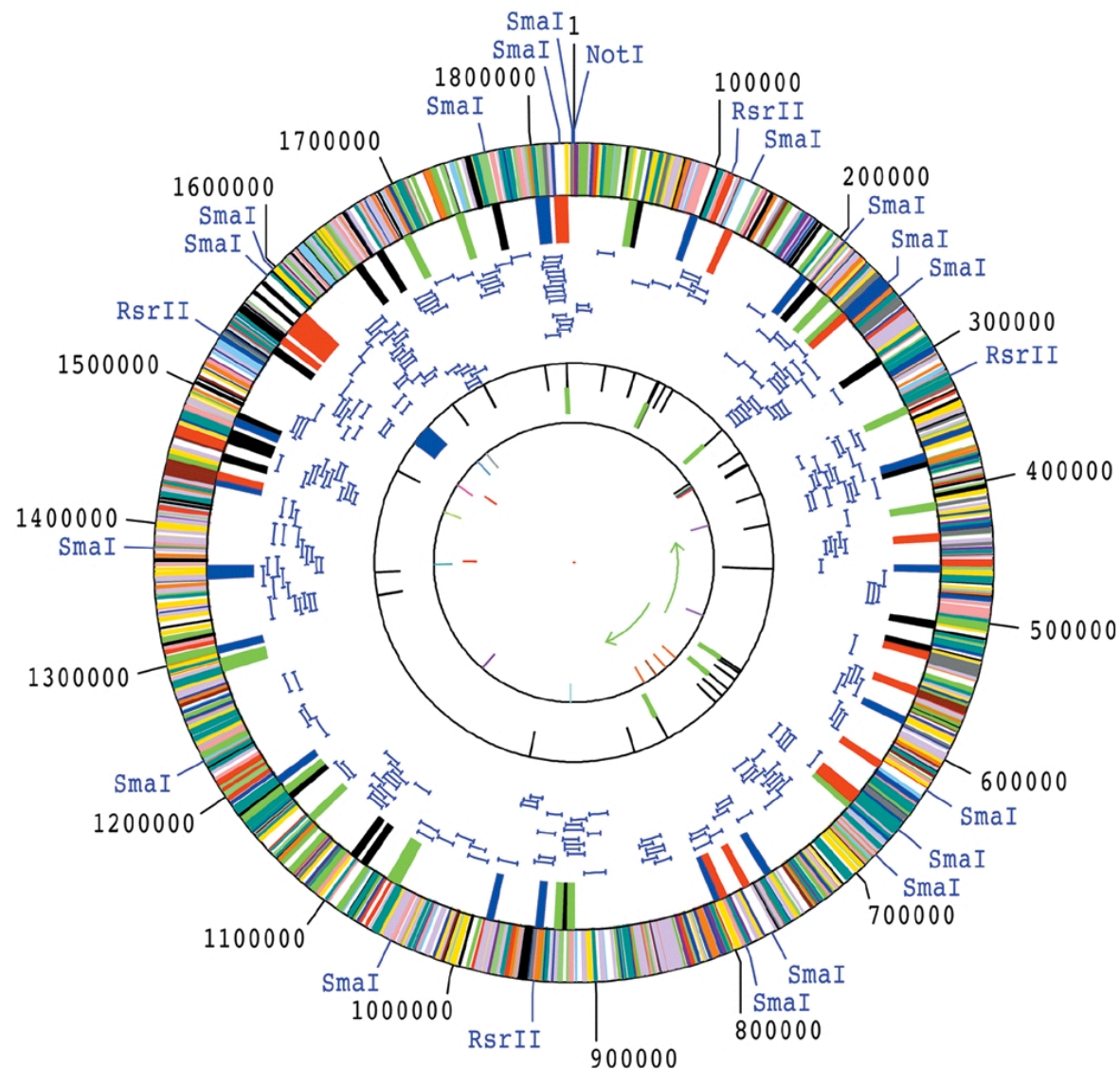


Fluorescence detection of oligonucleotide fragments produced by dideoxy method

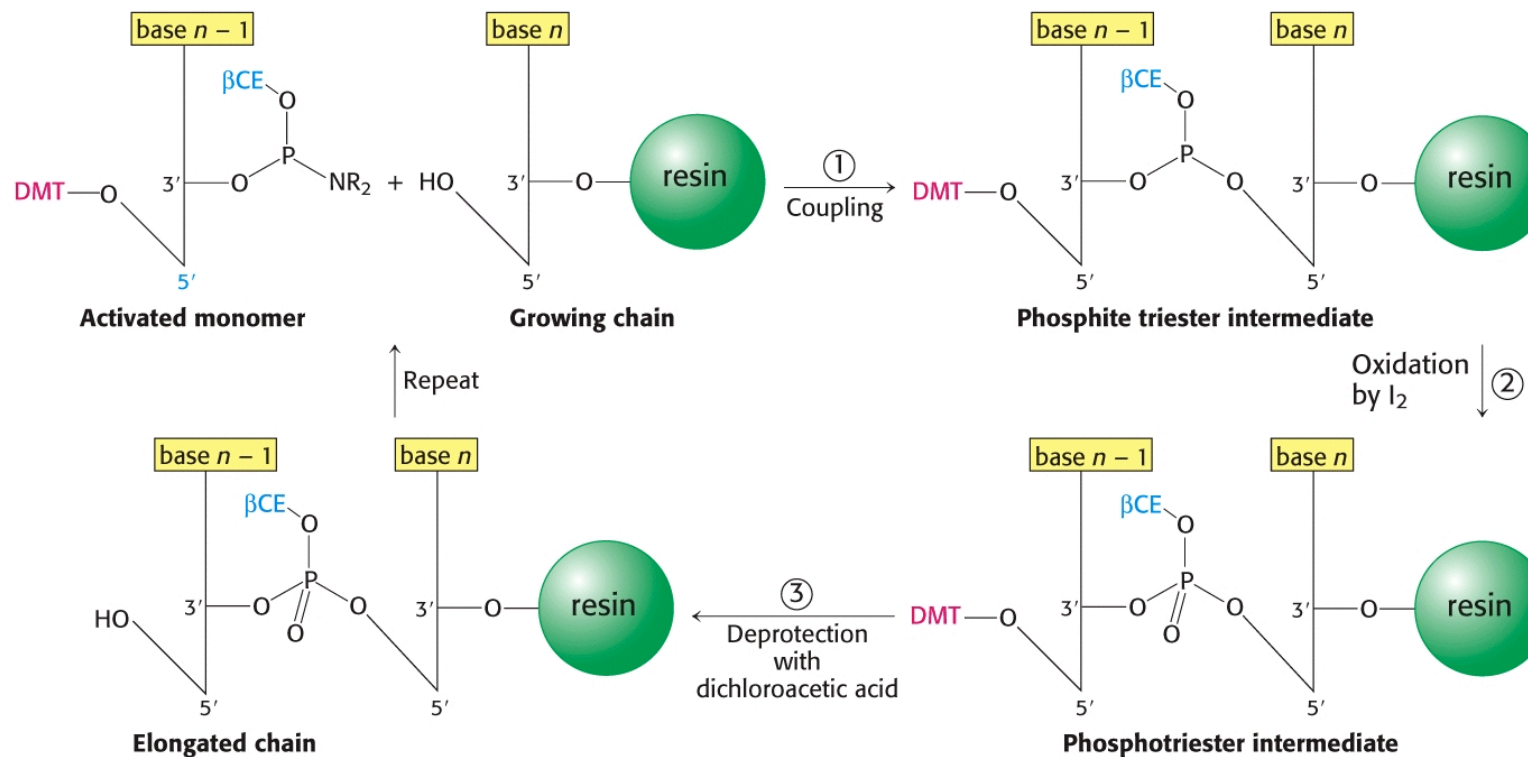


Complete genome (Haemophilus influenza)

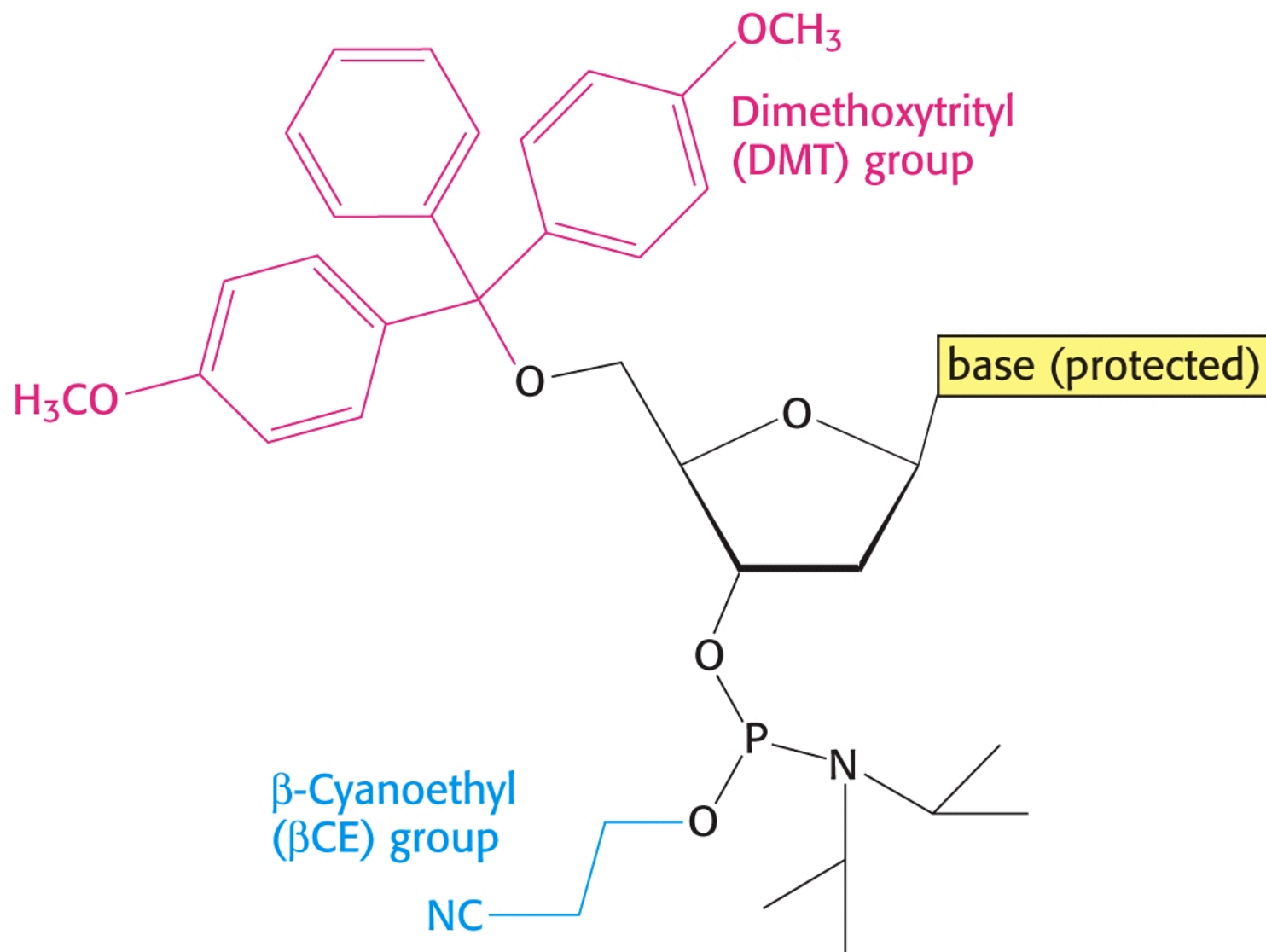
First sequenced genome of free living organism



Chemical synthesis of DNA probes and genes



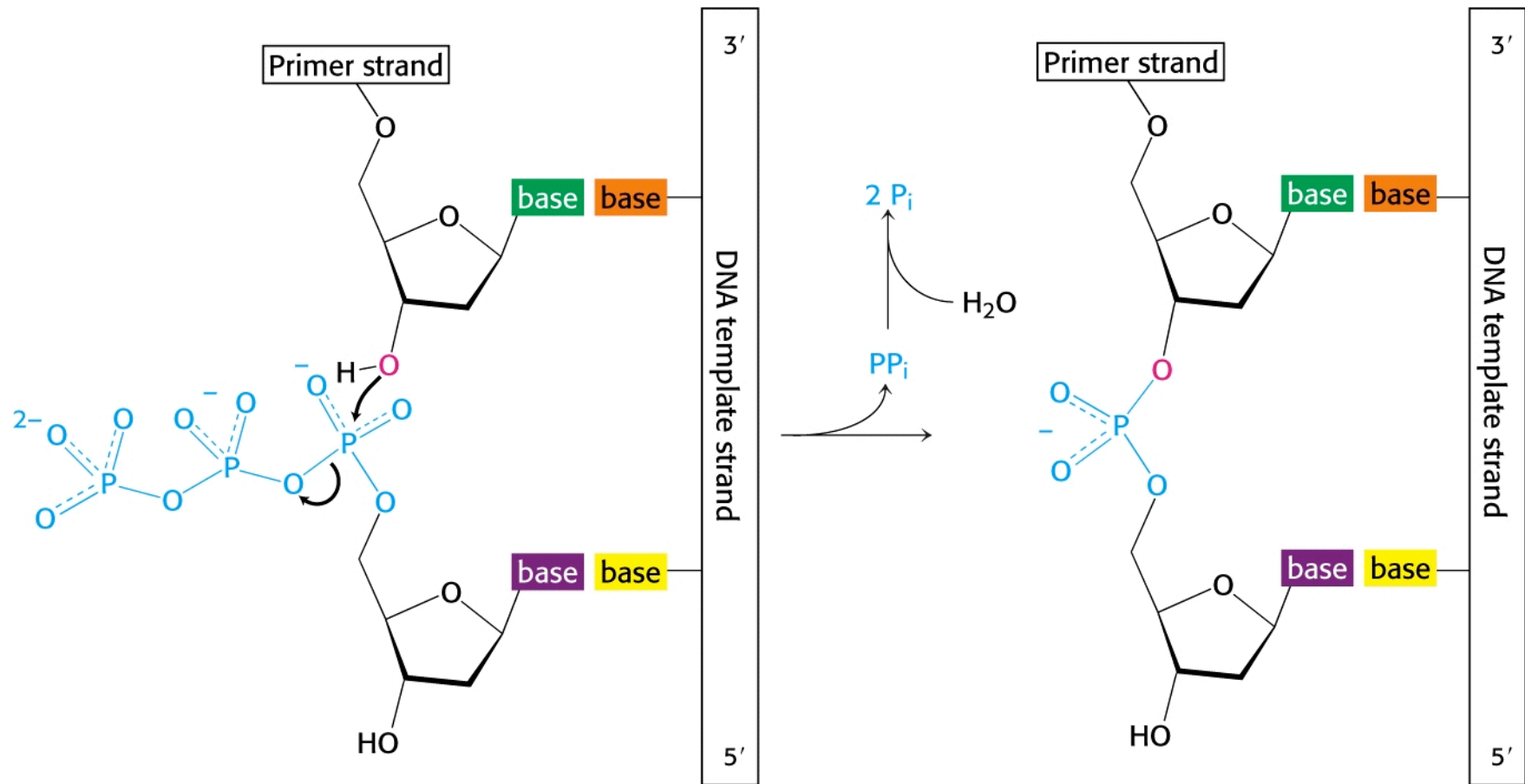
DNA strands like polypeptides can be synthesized by the sequential addition of activated monomers to a growing chain that is linked to an insoluble support



**A deoxyribonucleoside 3'-phosphoramidite
with DMT and β CE attached**

DNA replication by DNA polymerase

(a template directed enzyme)



Polymerase Chain Reaction (PCR)

Kary Mullis, 1984

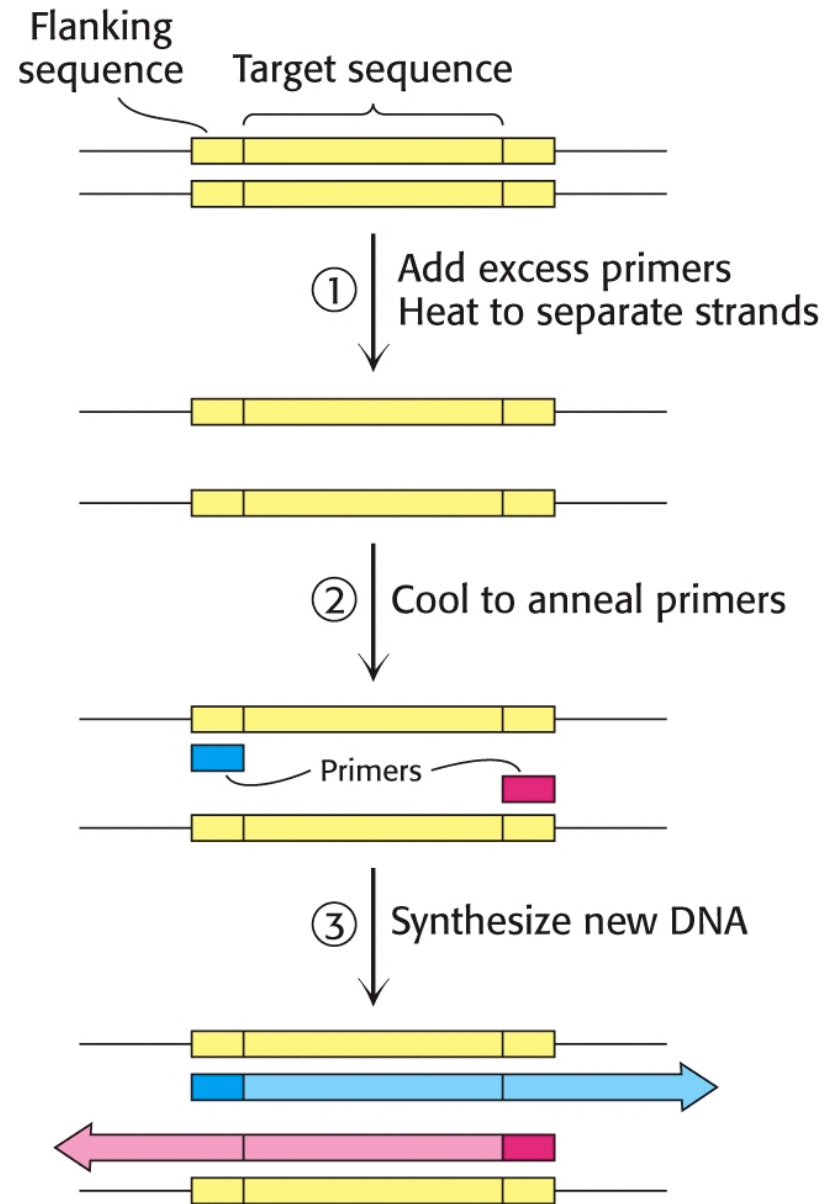
**Selected DNA sequences
can be greatly amplified by PCR**

Add:

1. Primers
2. dNTPs
3. Heat- DNA polymerase

First Cycle of PCR

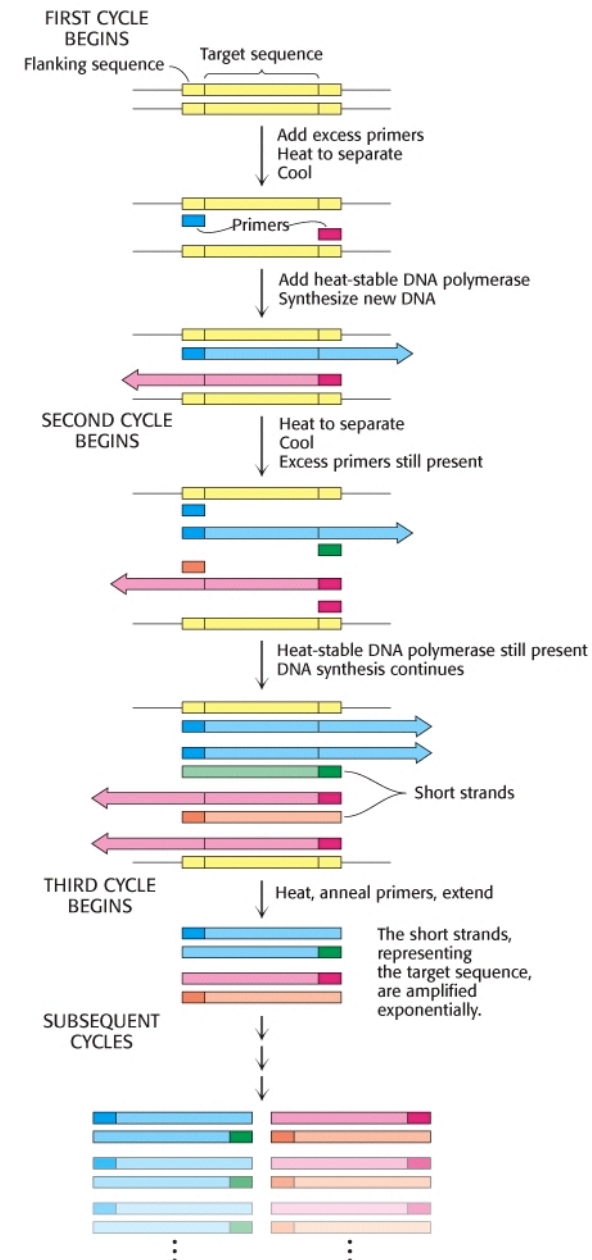
1. Strand separation
2. Hybridization of primers
3. Extension by DNA synthesis



Multiple cycles of PCR

Short target sequences at the end of 3rd cycle are subsequently amplified exponentially, parent sequences arithmetically

A single DNA molecule can be amplified and detected



Application of PCR

- 1 DNA molecule can be amplified
- Medical diagnostics (bacterial & viral infection e.g. HIV, tuberculosis, cancer)
- Forensics, legal medicine
- Analyzing ancient DNA

Recombinant DNA technology has revolutionized biology

New combinations of unrelated genes can be constructed

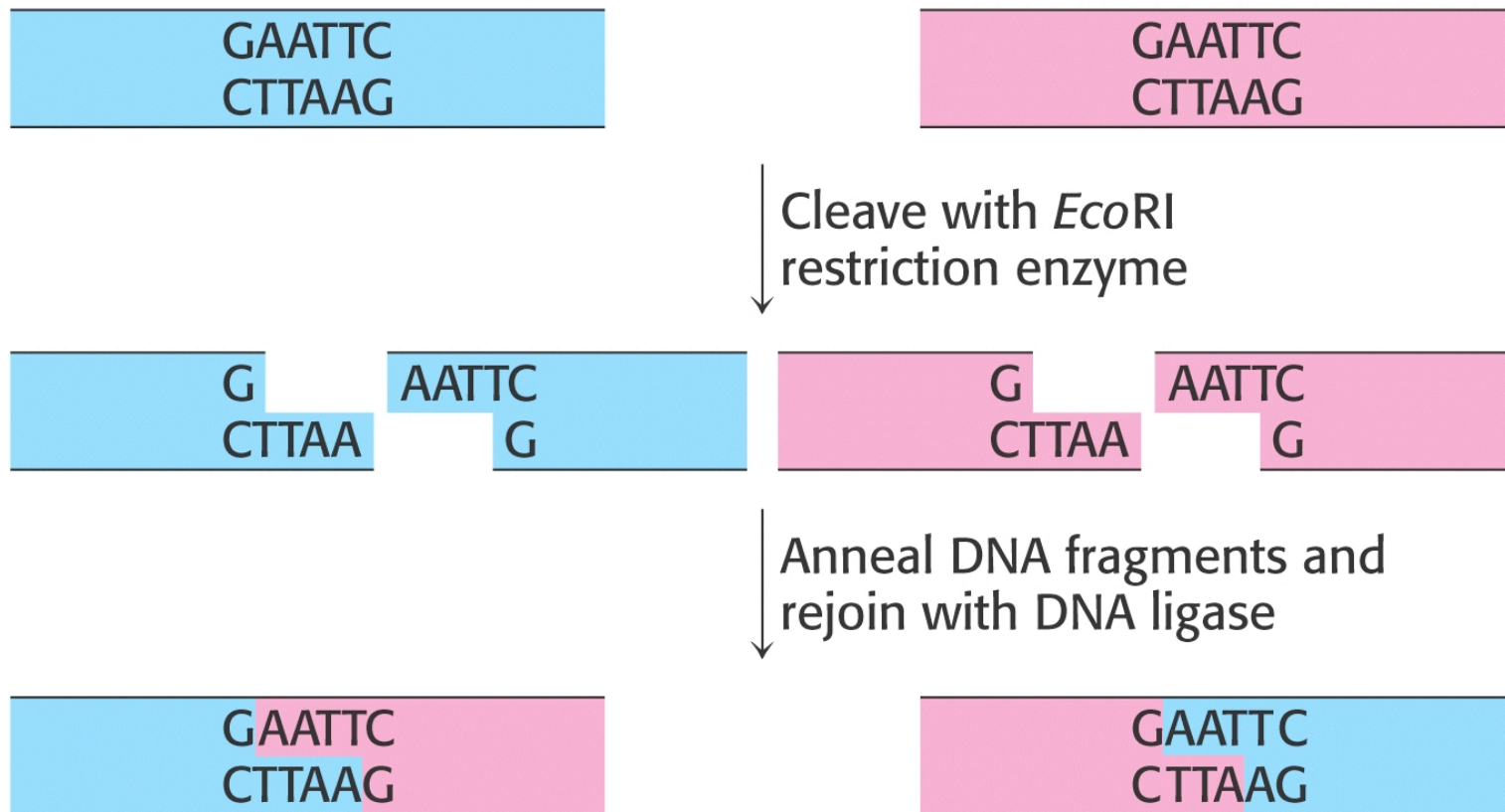
These novel combinations can be cloned = amplified manifold by introducing them into suitable cells where they are replicated by DNA-synthesizing machinery of the host

How can novel DNA molecules be constructed?

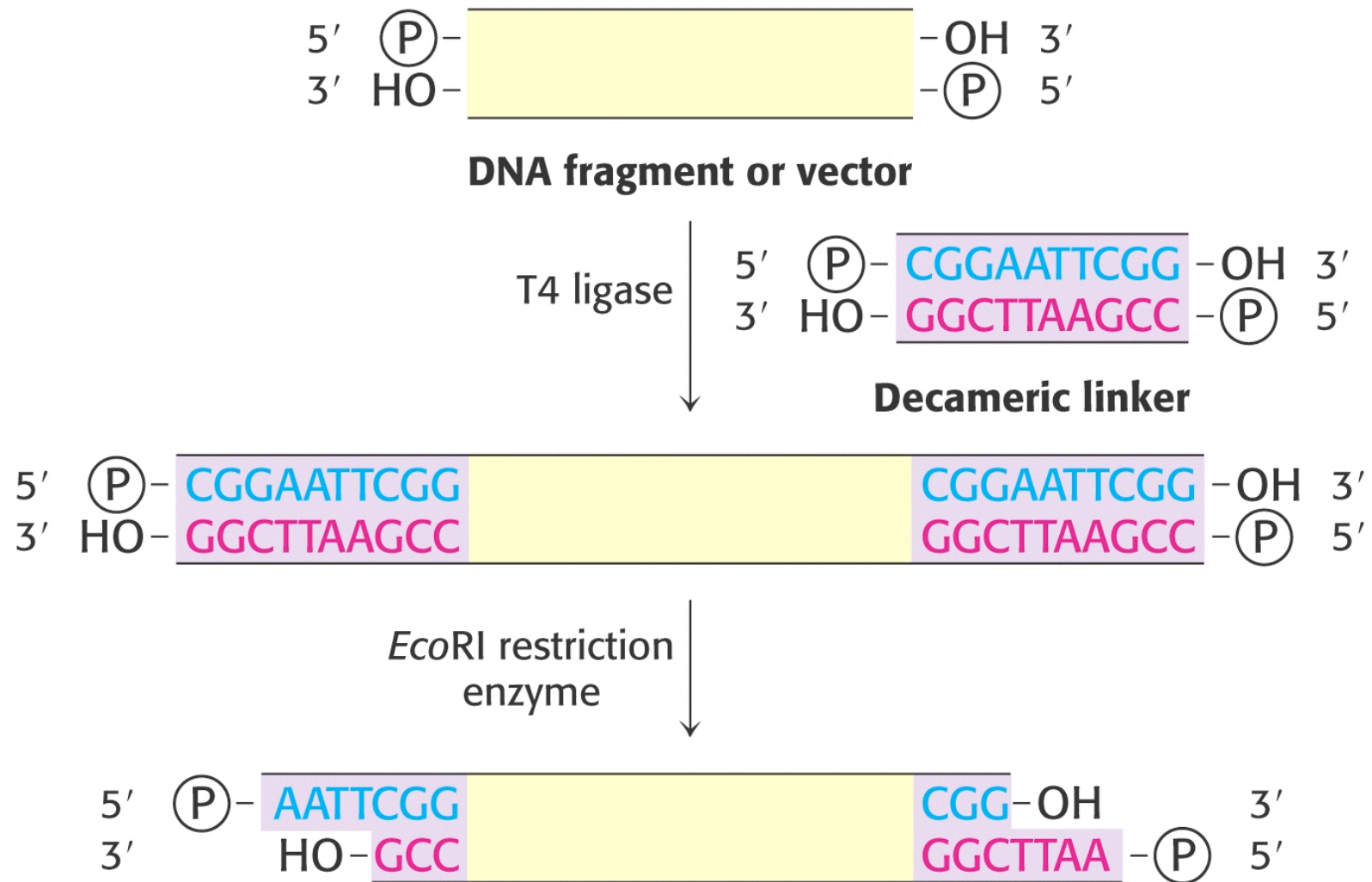
- DNA fragment of interest is covalently joined to a DNA **vector**
- A **vector** can replicate autonomously in an appropriate host
- **Plasmids** (circles of DNA in bacteria) and **bacteriophages** are vectors of choice for cloning in *E.coli*

Joining DNA molecules by the cohesive-end method

Two DNA molecules are cleaved with a restriction enzyme and then are ligated to form new, recombinant molecules



Formation of cohesive ends by addition and cleavage of a chemically synthesized linker

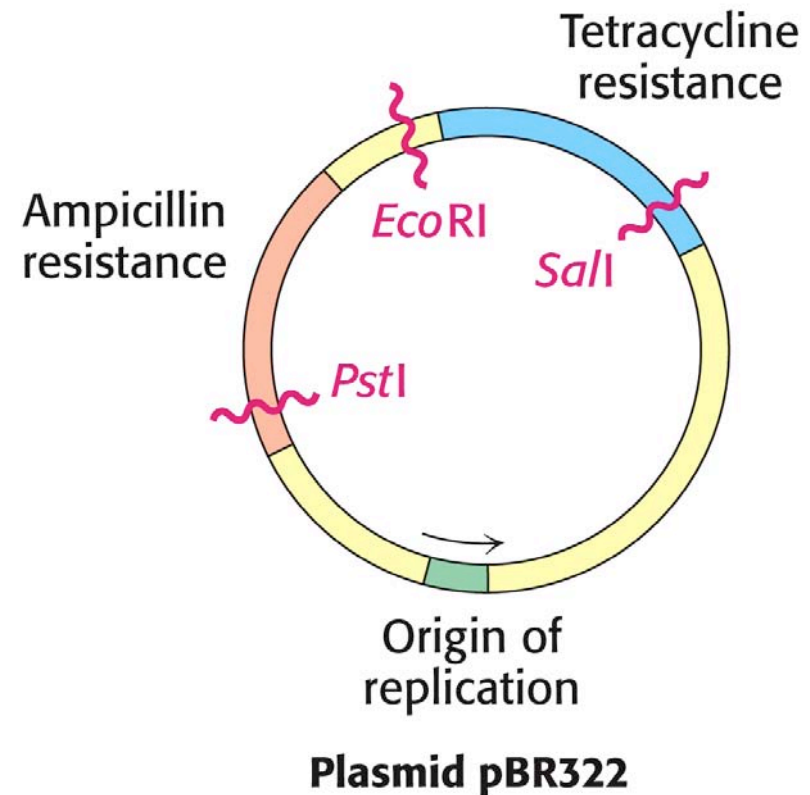


Plasmids are choice vectors for DNA cloning in bacteria

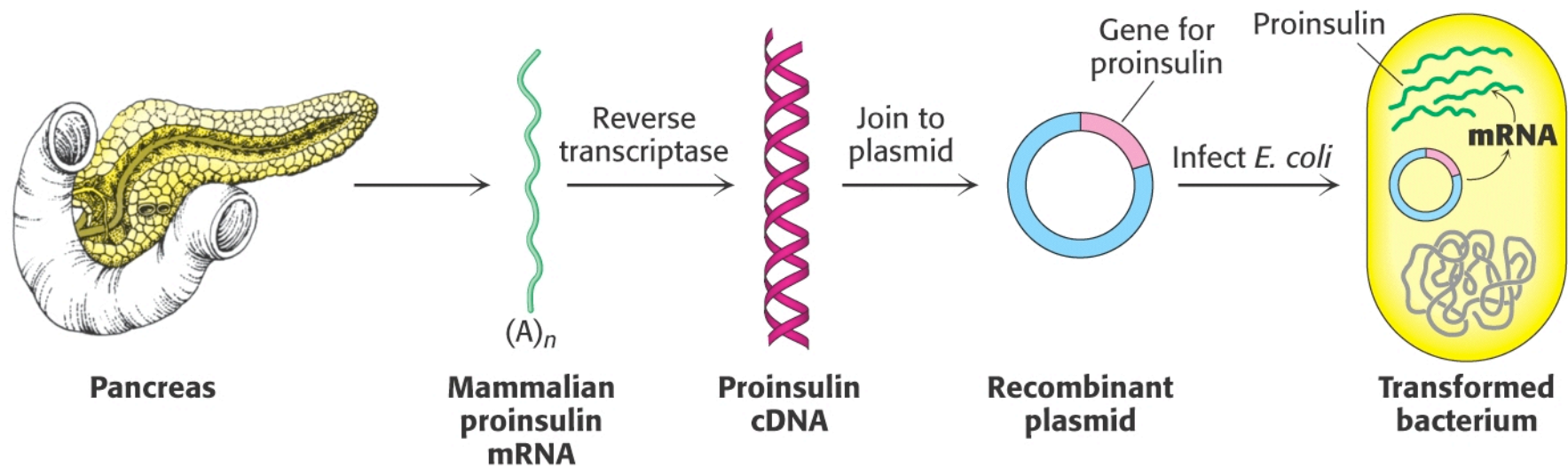
Circular duplex DNA in some bacteria

Carry genes for inactivation of antibiotics

Can replicate independently of host chromosome

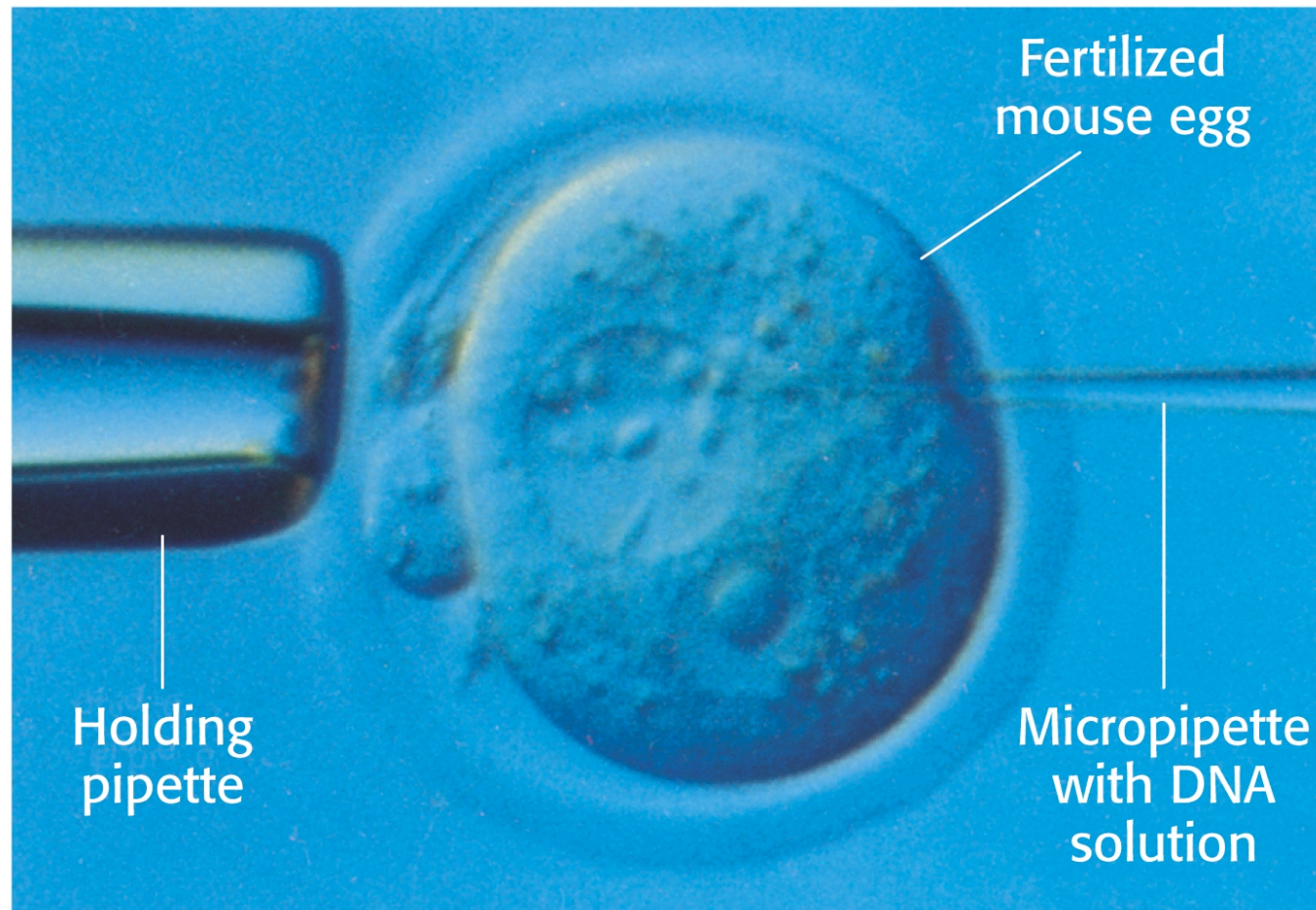


Synthesis of proinsulin by bacteria



New genes can be inserted into eukaryotic cells
e.g. by microinjection of DNA

Why? To receive posttranslational modifications



Summary

- **Basic tools:** Restriction enzymes, detection of DNA fragments, rapid sequencing, DNA synthesis, PCR
- **Recombinant DNA technology:** Construction of novel DNA molecules, recombinant production of proteins
- **Manipulating genes in eukaryotes**
- **Engineering proteins:** Site specific mutagenesis